Bike number crunch

How Bike Derailleurs Work

Name _____ Date _____

Write one answer in each blank. Show how you traded ones and tens when you regrouped.

1. One parts tray holds 28 gear teeth cutouts and another holds 35. How many teeth cutouts are there in all?

28 + 35 = _____

Show your thinking.

2. A rider plans 72 pedal turns and has finished 38. How many pedal turns remain?

72 − 38 = _____

Show your thinking.

3. Two ride notes record 47 wheel turns and 18 more wheel turns. How many wheel turns do they record altogether?

47 + 18 = _____

Show your thinking.

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4. A box begins with 61 gear-number cards and 24 are used. How many gear-number cards are left?

61 − 24 = _____

Show your thinking.

5. One bike outing lasts 56 minutes and another lasts 28 minutes. How many minutes is that in all?

56 + 28 = _____

Show your thinking.

6. A gear maker checks 93 teeth and marks 47 as finished. How many teeth still need checking?

93 − 47 = _____

Show your thinking.

A loop with teeth

The chain loops around a front chainring and one back cog. When the pedals turn forward, the moving chain helps turn the rear wheel.

Cable and spring

A shifter pulls or releases a cable. The rear derailleur swings sideways and guides the moving chain toward the next back cog.

Two pulley jobs

The upper pulley guides the chain. The lower pulley and a spring take up extra chain, keeping the long loop snug.

The front cage

On bikes with several front chainrings, a metal cage nudges the moving chain sideways from one chainring to another.

Keep it moving

The chain must move forward to climb onto another cog. A shifter click alone cannot slide a still chain into place.

Easy or hard

A big back cog makes climbing easier. A small back cog turns the wheel farther per pedal turn but needs more push.

Make a little book

1 Fold it into 8 boxes. Open it flat.

2 Fold short end to short end. Cut the thick line. Open it.

3 Fold the long way, pictures out. Push the ends in.

4 Fold into a book. Page 1 goes on the front!

WONDER DAILY

How Bike Derailleurs Work

Write the path of a shift

How Bike Derailleurs Work · Curiosity day

Name _____ Date _____

Choose one clue. Say the whole idea, then draw or write it.



A sideways nudge



Two little wheels



Easy gear, hard gear

Draw one bike gear clue and add a small arrow to the part whose job you explain.

gear · chain · cog · teeth · derailleur · cable · spring · lever · shift · easy gear · hard gear

When the lever pulls the cable, … dictate what the rear derailleur does

The lower pulley keeps the chain snug by … dictate how it takes up slack

A big back cog feels easier because … dictate what changes for each pedal turn

Does your sentence name a part, tell what it does, and connect that job to the chain?